Date: 17.09.2024

III SEMESTER B.Tech (IT, ITNS) | Roll No.________

MID-SEMESTER EXAMINATION 2024

Course Code : INECC 305/ITECC305

Course Title : Probability & Stochastic Processes

Time: 1:30 Hrs

Max Marks: 20

Note: -For Re-registration ie **INECC06 / EIECC05, marks shall be scaled up to 25 marks.** *Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.*

Q. No.		Questions	Marks	CO
Q 1	a	Let A and B be events with probabilities P[A] and P[B] a. Find $P[A \cup B]$ if A and B are independent b. Find $P[A \cup B]$ if A and B are mutually exclusive. c. Show that $P[A \cap B \cap C] = P[A \mid B \cap C]P[B \mid C]P[C]$	03	CO1
	b	Consider the two random variables (X, Y). Find the region of the plane corresponding to the events $A = \{X + Y \leq 10\}, B = \{\min(X, Y) \leq 5\}$, and $C = \{X^2 + Y^2 \leq 100\}$.	01	CO2
Q 2	a	Two friends decide to meet at a certain location. If each of them independently arrives at a time uniformly distributed between 12 noon and 1 P.M., find the probability that the first to arrive has to wait longer than 10 minutes	03	CO1
	b	The random variable X has CDF $$F_X(x) = \begin{cases} 0 & x < -3 \\ 0.4 & -3 \leq x < 5 \\ 0.8 & 5 \leq x < 7 \\ 1 & x \geq 7 \end{cases}$$ Write $P_X(x)$, the PMF of X.	01	CO2
Q 3	a	Let X be a random variable with pdf $$f_X(x) = \begin{cases} 0 & x < 0 \\ ce^{-2x} & x \geq 0 \end{cases} \quad (c > 0)$$ (a) Find c (b) Let a > 0, x > 0, Find $P[X \geq x + a]$ (c) Let a > 0, x > 0, Find $P[X \geq x + a \mid X \geq a]$	03	CO2
	b	Compute $E[\lvert X\rvert^n]$, if X is Uniformally distributed with zero mean and variance equal to 2.	01	CO2
Q 4	a	The joint density function of X and Y is given by: $$f_{X,Y}(x, y) = \begin{cases} 2e^{-x}e^{-y} & 0 < x < \infty, 0 < y < \infty \\ 0 & otherwise \end{cases}$$ Compute (i) $P\{X > 1, Y < 1\}$ (ii)$P\{X < Y\}$(iii)$P\{X < a\}$ (ii) $f_X(x)$	03	CO2
	b	What are the properties of Joint CDF, Joint PDF and Joint PMF.	01	CO1
Q 5	a	Suppose the joint PMF of a bivariete r.v. (X,Y) is given by: $$p_{XY}(x_i, y_j) = \begin{cases} \frac{1}{3} & (0,10), (1,0), (2,1) \\ 0 & otherwise \end{cases}$$ (i) Are X and Y independent? (ii) Are X and Y uncorrelated?	03	CO3
	b	Random variable X has mean 'm' and variance σ^2. Compute the mean of Y=$2Y = 2X^2 + 3X - 4$	01	CO3